Zoledronic acid

Zoledronic acid is used to treat and prevent problems caused by the effect of cancer on the bones.

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What is zoledronic acid?

Zoledronic acid belongs to a group of drugs called bisphosphonates. It can be used to:

- reduce high calcium levels in the blood (hypercalcaemia) that can be caused by cancer
- prevent bone weakness or bone fractures caused by myeloma or cancers that have spread to the bones (advanced cancers)
- prevent bone thinning and osteoporosis caused by treatments used for early-stage breast cancer
- reduce the risk of breast cancer spreading to the bones in early-stage breast cancer.

Cancer that has spread to the bones is called secondary bone cancer. It happens when cells from the original (primary) cancer spread to form a new tumour in the bone. This is sometimes called secondary cancer or metastasis.

It is best to read this information with our general information about bisphosphonates and the type of cancer you have.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the electronic Medicines Compendium (eMC) website, which has patient information leaflets (PIL) for individual drugs.

The effect of cancer on the bones

Bone cells constantly renew themselves. This helps bones keep their strength and shape.

There are 2 types of cell inside the bones. These are called:

- osteoclasts, which break down and remove old bone
- osteoblasts, which build new bone.

As we get older, osteoclasts begin to break down more bone than osteoblasts make to replace it. This means our bones slowly become thinner (less dense). In some people, too much of the bone density is lost and this increases the risk of osteoporosis and bone fractures.

Some secondary bone cancers make a substance that causes osteoclasts to destroy more bone than can be replaced. The affected bone becomes weak and painful and can break more easily.

Bones contain calcium, which gives them strength. A bone affected by cancer may cause calcium from the bones to be released into the blood. A raised level of calcium in the blood is called hypercalcaemia. This may cause symptoms including:

- feeling sick (nausea)
- being sick (vomiting)

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- feeling thirsty
- needing to pee (pass urine) more often
- constipation
- tiredness
- irritability
- confusion.

How zoledronic acid works

Zoledronic acid reduces the activity of the cells that break down the bone (osteoclasts). This can help reduce pain and strengthen the bone. For primary breast cancer, it can reduce the risk of cancer spreading to the bones.

Zoledronic acid also reduces the amount of calcium that is lost from the bones. This helps calcium levels in the blood return to normal.

How zoledronic acid is given

When you have your treatment, you will meet someone from your cancer team, such as a:

- cancer doctor
- specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

During the course of your treatment, you will have regular blood tests. This is to check that it is safe for you to have treatment.

You will meet with a doctor, nurse or pharmacist before you have treatment. They will talk to you about your blood results and ask how you have been feeling.

Zoledronic acid can be given on its own or with other cancer treatments.

You have zoledronic acid by a drip (infusion) into the vein. You usually have it in the outpatient department at the hospital. The infusion takes at least 15 minutes. It is usually given once every 3 to 4 weeks.

You may have it through 1 of the following:

- a cannula – a short, thin tube the nurse puts into a vein in your arm or hand
- a central line – a fine tube that goes under the skin of your chest and into a vein close by
- a PICC line – a fine tube that is put into a vein in your arm and goes up into a vein in your chest
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an implantable port (portacath) – a disc that is put under the skin on your chest or arm and goes into a vein in your chest.

Your course of zoledronic acid

How long zoledronic acid is given for depends on why you are having the treatment.

- If you are having zoledronic acid to help prevent cancer coming back, you will have an infusion once every 6 months for 3 to 5 years.
- If you are having zoledronic acid to reduce pain or strengthen your bones, you usually have an infusion every 3 to 4 weeks. You may also need to take calcium and vitamin D supplements. You usually need to take zoledronic acid for at least 2 to 3 months for it to have the full effect on your bones. After that, you can usually take it for as long as it is working well.
- If you are having zoledronic acid to lower calcium levels in your blood, it is usually given as a single dose.

Your doctor will tell you more about your treatment and how long you may need to have it.

Dental check-up before treatment

Rarely, this treatment can affect the jawbone. You can read more about this in the side effects section below.

Keeping to a regular oral hygiene routine keeps your mouth and teeth healthy. This reduces your risk of jaw problems. You usually need a dental check-up to check for any problems before you start treatment.

If you need urgent treatment to lower high calcium levels in your blood, you do not usually need to have a dental check-up first.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Other cancer treatments may cause different side effects. If you are also having other cancer treatment, you may have other side effects.

Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

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It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

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Side effects while being given

Allergic reaction

Some people have an allergic reaction while having this treatment. Signs of a reaction can include:

- feeling hot or flushed
- shivering
- itching
- a skin rash
- feeling dizzy or sick
- a headache
- feeling breathless or wheezy
- swelling of your face or mouth
- pain in your back, tummy or chest.

Your nurse will check you for signs of a reaction during your treatment. If you feel unwell or have any of these signs, tell them straight away. If you do have a reaction, they can treat it quickly.

Sometimes a reaction happens a few hours after treatment. If you develop any of these signs or feel unwell after you get home, contact the hospital straight away on the 24-hour number.

Always call 999 if swelling happens suddenly or you are struggling to breathe.

Pain along the vein

This treatment can cause pain:

- at the place where the drip (infusion) is given
- along the vein.

If you feel pain, tell your nurse straight away. They can check the site. They may give the drug more slowly or flush it through with more fluid to reduce pain.

Possible side effects

Flu-like symptoms

Some people have flu-like symptoms, such as:

- a high temperature
- chills
- fatigue (tiredness)
- pains in your muscles or joints.

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Tell your doctor if these symptoms do not get better within 2 days or are getting worse. It may help to take mild painkillers, such as paracetamol. Contact the hospital on the number you have been given. Your doctor, nurse or pharmacist can give you advice.

Feeling tired (fatigue)

Feeling tired is a common side effect. Try to pace yourself and plan your day so you have time to rest between activities.

Being physically active can help to manage tiredness and give you more energy. It also:

- helps you sleep better
- reduces stress
- improves your bone health.

If you feel sleepy, do not drive or operate machinery.

Pain in the thigh, hip or groin

Sometimes pain in muscles or bones gets worse for a short time when you start taking this treatment. If this happens, your doctor can prescribe painkillers for you until it gets better.

Rarely, people having this treatment develop a break (fracture) in their thigh bone without an obvious cause. Sometimes both thigh bones are affected.

If you have any thigh, hip or groin pain, tell your doctor. Let them know that you are taking bisphosphonates. They can arrange tests to check the thigh bones for any signs of bone weakness or fracture.

Anaemia (low number of red blood cells)

This treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia. You may feel:

- very low in energy
- breathless
- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

Headaches and dizziness

You may have headaches or feel dizzy at times when taking this treatment. Tell your doctor if you notice these symptoms.

Effects on the eyes

This treatment can cause eye problems. If you have any of these symptoms during treatment or after it finishes, contact the hospital on the 24-hour number straight away. Contact them if:

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- you notice any change in your vision, such as double vision or things appear blurry or faded
- your eyes get red and inflamed (conjunctivitis)
- you have pain in your eyes
- your eyes are sensitive to light

You may need to have an eye check with a specialist eye doctor (ophthalmologist).

Feeling sick (nausea)

You may feel sick or be sick (vomit) while having this treatment. This is usually mild. Your doctor can prescribe anti-sickness drugs to help prevent or control sickness.

Take the drugs exactly as your doctor, nurse or pharmacist tells you to. It is easier to prevent sickness than treat it after it has started.

Loss of appetite

This treatment can affect your appetite. Don’t worry if you do not eat much for 1 or 2 days. But if your appetite does not come back after a few days, or if you are losing weight, tell your doctor, nurse or pharmacist. They can give you advice. They may give you food or drink supplements. Or they may suggest changes to your diet or eating habits to help.

Effects on the kidneys

This treatment can affect how the kidneys work. This is usually mild and goes back to normal after treatment ends. You will have blood tests to check how well your kidneys are working. Contact the hospital on the 24-hour number if you:

- have blood in your urine (pee)
- are passing less urine or peeing less often than usual.

Drinking fluids helps protect your kidneys. The advice is usually to try to drink at least 2 litres (3½ pints) of fluid each day. But follow any advice from your doctor, nurse or pharmacist about how much is right for you.

Numbness or tingling

You may notice numbness or tingling around the mouth, or in the fingers and toes. Contact your doctor or the hospital straight away if you have these symptoms. This may be caused by low levels of calcium in your blood. You will have regular blood tests to check your calcium levels.

Your doctor may ask you to take calcium and vitamin D supplements, unless you are having this treatment to lower the levels of calcium in your blood. Your doctor will let you know if you need any supplements.

Effects on the heart

This treatment can affect how the heart works. You may have tests to check how well your heart is working. These may be done before, during and after treatment.

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If the treatment is causing heart problems, your doctor may change the type of treatment you are having.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- breathlessness
- dizziness
- changes to your heartbeat (palpitations)
- swollen feet and ankles.

Other conditions can cause these symptoms, but it is important to get them checked by a doctor.

Always call **999** if you have:

- chest pain, pressure, heaviness, tightness or squeezing across the chest
- difficulty breathing.

Diarrhoea

This treatment may cause diarrhoea. Diarrhoea means passing more stools (poo) than is normal for you, or having watery or loose stools. You may also have stomach cramps. If you have a stoma, it may be more active than usual.

If you are passing loose stools 3 or more times a day and this is not normal for you, contact the hospital as soon as possible on the 24-hour number. Follow the advice they give you about:

- taking anti-diarrhoea medicines
- drinking enough fluids to keep you hydrated and to replace lost salts and minerals
- any changes to your diet that might help.

They might also ask you for a specimen of your stool to check for infection.

Changes to your taste

Some foods may taste different or have no taste. Try different foods to find out what tastes best to you. You may also get a bitter or metallic taste in your mouth. Your doctor, nurse or pharmacist can give you advice. It might help to try:

- sucking sugar-free sour or boiled sweets
- eating cold foods
- eating sharp-tasting fresh fruit.

Taste changes usually get better after treatment ends. We have more information about coping with changes to taste.

Tummy pain

You may get pain in your tummy (abdomen), or have indigestion. Your doctor, nurse or pharmacist can give you advice or treatment to help. Contact the hospital straight away on the 24-hour number if your symptoms do not go away, or get worse.

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Skin changes

This treatment can affect your skin. It might feel dry. You may develop a rash, which may be itchy.

Always tell your doctor, nurse or pharmacist about any skin changes. They can give you advice or prescribe creams or medicines to help.

If your skin feels dry, try using soap-free cleansers and unperfumed moisturising cream every day.

Effects on the lungs

This treatment can cause changes to the lungs. Tell your doctor, nurse or pharmacist if you develop:

- a cough that does not go away
- wheezing
- breathlessness.

You should also tell them if any existing breathing problems get worse. You may have tests to check your lungs.

Jaw problems

Rarely, this treatment can affect the jawbone. Healthy bone in the jaw becomes damaged and dies. This is called osteonecrosis of the jaw (ONJ). It can cause:

- pain
- loosening of the teeth
- problems with the way the gums heal.

The risk of jaw problems is higher after some types of dental treatment. It is also higher in people who smoke, have gum disease or dentures that do not fit well.

It is important to avoid having dental treatment that could affect your jawbone when you are having bisphosphonates. This includes having a tooth or root removed or dental implants put in. You can still have fillings, gum treatments or a scale and polish.

To reduce your risk of developing jaw problems, your doctor will advise you to:

- have a full dental check-up before starting treatment
- look after your teeth and gums during treatment (ask your dentist for advice)
- tell your dentist you are taking a bisphosphonate before having any dental treatment.

Tell your cancer specialist and dentist straight away if you at any time you develop:

- pain
- swelling or redness in your gums
- numbness or heaviness in your jaw
- loose teeth.

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Ear problems

Very rarely, bones in the outer ear may be affected by this treatment. Always tell your doctor if you have any:

- ear pain
- discharge from your ear
- ear infections.

You should also let them know if you notice any other changes in your ears or hearing.

Other information

Going into hospital

If you are admitted to hospital for a reason not related to the cancer, it is important to tell the doctors and nurses that you are having treatment with bisphosphonates. You should tell them the name of your cancer specialist. This is so they can talk to them for advice.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- herbal drugs and complementary or homeopathic therapies
- recreational drugs – for example, cannabis.

Breastfeeding

You are advised not to breastfeed while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

Contraception

Your doctor, nurse or pharmacist will advise you not to get pregnant or make someone pregnant while having this treatment and for some time afterwards. If you are pregnant, think you might be pregnant or are planning to have a baby, you must tell your doctor as soon as possible. The drugs may harm a developing baby. It is important to use contraception to prevent pregnancy. Follow their advice about:

- what types of contraception to use
- how long after treatment you should continue to use contraception.

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Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References

Visit the electronic Medicines Compendium (eMC) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

Cancer Information Development team

Author

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Our cancer information meets the PIF TICK quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about how we produce our information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
-

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use illustrations to explain text

- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as ‘you’ so that everyone feels included. Where clinically necessary we use the terms ‘men’ and ‘women’ or ‘male’ and ‘female’. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information here.

How we can help

How to print a PDF from this website

Instructions on how you can make a PDF version of this page using the print function on your browser.

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don’t want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.



Cancer information in other formats and languages

We provide information in different languages and other formats, including audiobooks, Braille, BSL, easy read booklets. You can also request information in any language or format.

Get help

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